

Q1. what are the different protocols you observe at the following layers of the protocol stack?

Ans	Application layer	Transport layer	Network layer.
	MDNS	UDP	ARP
	NBNS	TCP	IPV6 ICMPv6
	DHCP		
	DNS		
	HTTP		

Q2 What is the total amount of data being received for the following two cases?

A) when you access:- <http://neverssl.com/>

B. when you access:- <https://www.cornell.edu/>

ip.addr==10.85.17.125

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.85.17.125	255.255.255.255	UDP	867	42981 → 29810 Len=825
365	10.037985195	10.85.17.125	255.255.255.255	UDP	867	42981 → 29810 Len=825
1156	30.111735919	10.85.17.125	255.255.255.255	UDP	867	42981 → 29810 Len=825

ip.addr==10.85.22.64									
No.	Time	Source	Destination	Protocol	Length	Info			
1	0.0000000000	10.85.22.64	224.0.0.251	MDNS	119	Standard query 0x000c PTR _googlecast._tcp.local, "QM" question PTR ...			
707	19.973533303	10.85.22.64	224.0.0.251	MDNS	119	Standard query 0x000d PTR _googlecast._tcp.local, "QM" question PTR ...			
1398	40.041847348	10.85.22.64	224.0.0.251	MDNS	119	Standard query 0x000e PTR _googlecast._tcp.local, "QM" question PTR ...			
2009	60.014450142	10.85.22.64	224.0.0.251	MDNS	119	Standard query 0x000f PTR _googlecast._tcp.local, "QM" question PTR ...			

Q.3

Answer the following when you access the site
http://neverssl.com/

A.) How many HTTP GET requests do you observe?
List down the GET requests.

Ans One HTTP GET requests observed.

B. GET requests: -

Time	Source	Destination	Protocol	length	Info
28.987171401	10.85.17.18	34.223.129.45	HTTP	142	GET/HTTP/1.1

B) What version of HTTP the server is running?
GET/HTTP/1.1

http.request.method=="GET"

No.	Time	Source	Destination	Protocol	Length	Info
1105	28.987171401	10.85.17.18	34.223.124.45	HTTP	142	GET / HTTP/1.1

c) For each of the HTTP GET requests, find out:

i) The total number of TCP segments being received

Ans One TCP segment received

ii) The total amount of data being received in the corresponding HTTP response message.

Ans 4261 bytes.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
1107	29.089349700	fe80::e45f:1bfff:fe4...	ff02::fb	MDNS	596	Standard query response 0x0000 PTR, cache flush Android_80X7M5NB.local A, cache flush 10.85.19.118 PTR, cache flush And...
1108	29.089360082	8a:26:d0:f4:f4:72	Broadcast	ARP	60	Gratuitous ARP for 10.85.22.163 (Reply)
1109	29.191440103	10.85.17.10	255.255.255.255	UDP	868	44246 -- 29810 Len=826
1110	29.191448414	10.85.22.43	255.255.255.255	UDP	867	47698 -- 29810 Len=825
1111	29.191467252	b0:19:21:74:04:80	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.17.10
1112	29.191467837	b0:19:21:74:0d:0a	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.22.43
1113	29.290422661	ASIXElectron_35:2d:...	Broadcast	ARP	60	Who has 10.85.23.1? Tell 10.85.16.1
1114	29.290423963	b0:19:21:74:0d:48	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.22.105
1115	29.290424107	b0:19:21:d4:8e:fa	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.16.145
1116	29.292714229	34.223.124.45	10.85.17.18	TCP	66	80 -- 34366 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3422388319 TSecr=2905658291
1117	29.292715891	34.223.124.45	10.85.17.18	HTTP	4327	HTTP/1.1 200 OK (text/html)
1118	29.292843925	10.85.17.18	34.223.124.45	TCP	66	34366 -- 80 [ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658597 TSecr=3422388320
1119	29.294132074	10.85.17.18	34.223.124.45	TCP	66	34366 -- 80 [FIN, ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658598 TSecr=3422388320
1120	29.399267293	10.85.17.205	255.255.255.255	UDP	868	57137 -- 29810 Len=826
1121	29.399265793	10.85.18.54	255.255.255.255	UDP	871	33504 -- 29810 Len=829
1122	29.399308352	b0:19:21:74:06:6e	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.18.54
1123	29.399308748	b0:19:21:d4:90:f8	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.17.205
1124	29.399308803	ASIXElectron_35:2d:...	Broadcast	ARP	60	Who has 10.85.18.83? Tell 10.85.16.1
1125	29.399308858	ASIXElectron_35:2d:...	Broadcast	ARP	60	Who has 10.85.20.231? Tell 10.85.16.1
1126	29.496981642	10.85.22.234	224.0.0.251	MDNS	82	Standard query 0x0001 PTR _googlecast._tcp.local, "QU" question
1127	29.496982349	fe80::2c6d:75fff:fe1...	ff02::fb	MDNS	102	Standard query 0x0001 PTR _googlecast._tcp.local, "QU" question
1128	29.496982474	10.85.22.234	224.0.0.251	MDNS	136	Standard query 0x0001 PTR _googlecast._tcp.local, "QU" question PTR %9E5E7C8F47989526C9BCD95D24084F60B27C5ED. sub. go...

Frame 1117: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlp0s20f3, 10.85.17.18 to 34.223.124.45
 Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: 44:38:e8:60:4d:a2 (44:38:e8:60:4d:a2)
 Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.17.18
 Transmission Control Protocol, Src Port: 80, Dst Port: 34366, Seq: 1, Ack: 77, Len: 4261

Source Port: 80
 Destination Port: 34366
 [Stream index: 0]
 [Conversation completeness: Complete, WITH_DATA (31)]
 [TCP Segment Len: 4261]
 Sequence Number: 1 (relative sequence number)
 Sequence Number (raw): 2373345713
 [Next Sequence Number: 4262 (relative sequence number)]
 Acknowledgment Number: 77 (relative ack number)
 Acknowledgment Number (raw): 2743240870
 1000 = Header length: 32 bytes (8)
 Flags: 0x018 (PSH, ACK)
 Window: 229
 [Calculated window size: 29312]
 [Window size scaling factor: 128]
 Checksum: 0xb3e [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
 [Timestamps]
 [SEQ/ACK analysis]
 [TCP payload (4261 bytes)]
 Hypertext Transfer Protocol
 Line-based text data: text/html (131 lines)

The TCP payload of this packet (tcp.payload), 4,261 bytes

```

0040 d3 b3 48 54 54 50 2f 31 2e 31 20 32 30 38 20 4f
0050 40 8d 0a 44 61 74 65 3a 20 54 68 75 2c 20 30 39
0060 20 4a 75 6c 20 32 30 32 30 20 30 37 3a 33 32 3a
0070 33 34 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20
0080 41 70 61 63 68 65 2f 32 2e 34 2e 36 36 20 28 29
0090 6d 0a 55 70 67 72 61 64 65 3a 20 68 32 2c 68 32
00a0 8d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 55
00b0 70 67 72 61 64 65 0d 0a 4c 61 73 74 2d 4d 6f 64
00c0 69 66 69 65 64 3a 20 57 65 64 2c 20 32 39 20 4a
00d0 75 6e 20 32 39 32 32 30 30 3a 32 33 3a 33 33
00e0 20 47 4d 54 0d 0a 54 61 67 3a 20 22 66 37 39
00f0 2d 35 65 32 38 62 32 39 64 33 38 65 39 33 22 0d
0100 8a 61 63 63 65 70 74 2d 52 61 6e 67 65 73 3a 20
0110 62 79 74 65 73 8d 0a 43 6f 6e 74 65 6e 74 2d 4c
0120 65 6e 67 74 68 3a 20 33 39 36 31 0d 0a 50 61 72
0130 79 3a 20 41 63 63 65 70 74 2d 45 6e 63 6f 64 69
0140 6e 67 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65
0150 3a 20 74 65 78 74 2f 68 74 6d 6c 3b 20 63 68 61
0160 72 73 65 74 3c 55 54 40 2d 38 0d 0a 0d 0a 3c 68
0170 74 6d 6c 3e 0a 89 3c 68 65 61 64 3e 0a 09 89 3c
0180 74 69 74 6c 65 3e 4e 65 70 62 73 53 4c 40 2d
0190 20 43 6f 6e 6e 65 63 74 69 6e 67 20 2e 2e 2e 20
01a0 3c 2f 74 69 74 6c 65 3e 0a 09 89 3c 73 74 79 6c
01b0 65 3e 0a 09 89 62 6f 64 79 20 7b 0a 09 09 89 66
01c0 6f 6e 74 2d 6e 61 6d 69 6c 79 3a 20 4d 6f 6e 74
01d0 73 65 72 62 71 74 2c 20 68 6e 6c 7b 65 74 69 63
01e0 61 2c 20 61 72 69 61 6c 2c 20 73 61 6e 73 2d 73
01f0 65 72 69 6b 3b 0a 09 09 66 6f 6e 74 2d 73 69
0200 7a 65 3a 20 31 36 7b 3b 0a 09 09 09 63 6f 6c 6f

```

Packets: 1379 - Displayed: 1379 (100.0%) - Dropped: 0 (0.0%)

Profile: Default

D) Identify TCP three way handshake used to establish the connection with the `http://neverssl.com`. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

Ans. SYN Packet:-

Source IP:- 10.85.17.18 (client)

Destination IP:- 34.223.124.45 (server)

Protocol info:- 57022 → 80 [SYN]

Sequence Number:- 0

• SYN-ACK Packet:-

Source IP:- 34.223.124.45 (server)

Destination IP:- 10.85.17.18 (client)

Protocol info:- 80 → 57022 [SYN, ACK]

Sequence Number:- 0
(ACK = 1)

• ACK Packet:-

Source IP:- 10.85.17.18 (client)

Destination IP:- 34.223.124.45 (server)

Protocol Info 57022 → 80 [ACK]

Sequence Number:- 1
(ACK = 1)

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp

No.	Time	Source	Destination	Protocol	Length	Info
918	24.572340111	10.85.17.18	34.223.124.45	TCP	74	34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905653876 TSecr=0 WS=1024
964	25.026736900	10.85.17.18	34.223.124.45	TCP	74	[TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905654931 TSecr=0 WS=1024
1004	26.05098197	10.85.17.18	34.223.124.45	TCP	74	[TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905655955 TSecr=0 WS=1024
1053	27.074717326	10.85.17.18	34.223.124.45	TCP	74	[TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905656979 TSecr=0 WS=1024
1084	28.098725545	10.85.17.18	34.223.124.45	TCP	74	[TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905658003 TSecr=0 WS=1024
1103	28.986746223	34.223.124.45	10.85.17.18	TCP	74	80 → 34366 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1460 SACK_PERM TSval=3422388026 TSecr=2905658003 WS=128
1104	28.986966889	10.85.17.18	34.223.124.45	TCP	66	34366 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2905658291 TSecr=3422388026
1105	28.987171401	10.85.17.18	34.223.124.45	HTTP	142	GET / HTTP/1.1
1116	29.292714229	34.223.124.45	10.85.17.18	TCP	66	80 → 34366 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3422388319 TSecr=2905658291
1117	29.292715891	34.223.124.45	10.85.17.18	HTTP	4327	HTTP/1.1 200 OK (text/html)
1118	29.292843925	10.85.17.18	34.223.124.45	TCP	66	34366 → 80 [ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658597 TSecr=3422388320
1119	29.294132074	10.85.17.18	34.223.124.45	TCP	66	34366 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658598 TSecr=3422388320
1134	29.606113857	34.223.124.45	10.85.17.18	TCP	66	80 → 34366 [FIN, ACK] Seq=4262 Ack=78 Win=29312 Len=0 TSval=3422388626 TSecr=2905658598
1135	29.606181896	10.85.17.18	34.223.124.45	TCP	66	34366 → 80 [ACK] Seq=78 Ack=4263 Win=60416 Len=0 TSval=2905658910 TSecr=3422388626

Frame 1117: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlp0s20f3, interface id 0
Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: 44:38:e8:60:4d:a2 (44:38:e8:60:4d:a2)
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.17.18
Transmission Control Protocol, Src Port: 80, Dst Port: 34366, Seq: 1, Ack: 77, Len: 4261
Hypertext Transfer Protocol
Line-based text data: text/html (131 lines)

```
0000  d3 b3 48 54 54 50 2f 31 2e 31 20 32 30 30 20 4f  HTTP/1.1 200 0
0001  4b 0d 0a 44 61 74 65 3a 20 54 68 75 2c 20 30 39  K. Date: Thu, 09
0002  20 4a 75 6c 20 32 30 32 36 20 30 37 3a 33 32 3a  Jul 202 6 07:32:
0003  33 34 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 2d 34 GMT- Server:
0004  41 70 61 63 68 65 2f 32 2e 34 2e 36 36 20 28 20 Apache/2.4.66 ()
0005  0d 0a 55 70 67 72 61 64 65 3a 20 68 32 2c 68 32  Upgrade: h2,h2
0006  63 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 55 c. Connec tion: U
0007  70 67 72 61 64 65 0d 0a 4c 61 73 74 2d 4d 6f 64 a pgrade. Last-Mod
0008  69 66 69 65 64 3a 20 57 65 64 2c 20 32 39 20 4a ified: Wed, 29 J
0009  75 6e 20 32 39 32 32 30 39 30 3a 32 33 3a 33 33 un 2022. 00:23:33
0010  20 47 4d 54 0d 0a 45 54 61 67 3a 20 22 66 37 39 GMT ET ag: "f79
0011  2d 35 65 32 38 62 32 39 64 33 38 65 39 33 22 0d -5e28b29 d38e93"
0012  0a 41 63 63 65 70 74 2d 52 61 6e 67 65 73 3a 20 Accept- Ranges:
0013  62 79 74 65 73 0d 0a 43 6f 6e 74 65 6e 74 2d 4c bytes -C ontent-L
0014  65 6e 67 74 68 3a 20 33 39 36 31 0d 0a 56 61 72 ength: 3 961 Var
0015  79 3a 20 41 63 63 65 70 74 2d 45 64 63 6f 64 69 y: Accept -Encodi
0016  6e 67 0d 0a 43 6f 6e 74 65 6e 74 2d 54 69 70 65 ng -Cont-ent-Type
0017  3a 20 74 65 78 74 2f 68 74 6d 6c 3b 20 63 68 61 : text/h tml; cha
0018  72 73 65 74 3d 55 54 46 2d 38 0d 0a 0d 0a 3c 68 rset=UTF -8...<h
0019  74 6d 6c 3e 0a 09 3c 68 65 61 64 3e 0a 09 09 3c tml> <h ead>...<
0020  74 69 74 6c 65 3e 4e 65 70 65 72 53 53 4d 20 2d title>Ne verSSL -
0021  20 43 6f 6e 65 63 74 69 6e 67 20 2e 2e 2e 20 Connect ing ...
0022  3c 2f 74 69 74 6c 65 3e 0a 09 09 3c 73 74 79 6c <title>...<styl
0023  65 3e 0a 09 09 62 6f 64 79 20 7b 0a 09 09 66 6e e>... bod y {...f
0024  6f 6e 74 2d 6e 61 6d 69 6e 79 3a 20 4d 6f 6e 74 ont-fami ly: Mon
0025  73 65 72 62 61 74 2c 20 68 65 6c 70 65 74 69 63 serrat, helvetic
0026  61 2c 20 61 72 69 61 6c 2c 20 73 61 6e 73 2d 73 a, arial , sans-s
0027  65 72 69 66 3b 0a 09 09 66 6f 6e 74 2d 73 69 erif; ...font-si
0028  7a 65 3a 20 31 36 78 3b 0a 09 09 09 63 6f 6c 6f ze: 16px; ...colo
```